Mahmoud Abd El-Ghaffar

AI Automation & Web Development Engineer

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Professional Summary

Computer Engineer and AI Automation Specialist combining hands-on expertise in full-stack website design and development with advanced systems integration. Experienced in designing visually compelling, mobile-first websites and web portals utilizing modern AI design workflows (Google Antigravity, Claude Code, rapid UI prototyping) alongside production business automations (Make, Zapier, n8n, CRMs, WhatsApp/Social funnels). Proven portfolio of live concept web design mockups and interactive prototypes deployed on Render (dental clinic mockups like Smile Inn, sovereign institutional portal concepts like FOE Procurement Portal, task assistant apps, and personal portfolio). Proven track record engineering 30+ native integration nodes in AI MicroMind Core and automating 30+ production workflows connecting CRMs (HubSpot, GoHighLevel, Airtable), forms (Typeform, Tally, Fillout), messaging platforms (WhatsApp Cloud API/WAPA, Instagram, Telegram, Slack), and databases. Comfortable working under strict NDA and proprietary data privacy standards.

Professional Experience

AI & Automation Engineer, AI MicroMind 2024 – 2026

- Built and co-built 30+ native integration nodes for WAPA (WhatsApp API), Telegram, Gmail, Weather, MCP Client, Pinecone, Webhooks, Schedulers, and PostgreSQL audit logging.
- Architected and deployed the AI MicroMind App with conversational BI and reporting capabilities, enabling natural-language access to operational data across Oracle, Microsoft Dynamics, SAP, Odoo, SQL, Oracle Fusion, and Dynamics 365 OData.
- Engineered responsive web applications and interactive client portals leveraging modern AI design workflows (Google Antigravity, Claude Code) for rapid UI iteration, clean UX, and component modularity.
- Developed AI agents and multi-step automation workflows using LangChain and LangGraph, including MCP-based integrations and backend execution flows.
- Built modular MicroApps and custom integrations connecting third-party business applications to the MicroMind ecosystem.
- Worked across deployment and infrastructure using Docker, AWS, DigitalOcean, Linux VPS, Render, and GitHub version-control workflows.

Automation Engineer, Operatives (IT Services) 06/2021 – 01/2022

Analyzed cross-department operational processes and built end-to-end automations to reduce repetitive manual work and data-entry bottlenecks.

Operation Engineer & Admin, Operatives (IT Services) 01/2020 – 06/2021

Managed IT operations, monitored workflow execution, supported operational tracking systems, and followed SLA-related processes.

Education

B.Sc. in Computer Engineering, Faculty of Engineering Shoubra, Benha University

Specialization: Computer Systems Architecture, Database Management Systems, Software Engineering & AI Systems.

Core Skills

Automation & AI Platforms
AI MicroMind, Flowise, n8n, Make, Zapier, LangChain, LangGraph

Website Design & UI/UX (AI-Assisted)
Responsive Web Design, Google Antigravity, Claude Code, Rapid UI Prototyping, Glassmorphism, Cross-Browser UI

AI, RAG & Data Engineering
AI Agents, Multi-Agent Workflows, Pinecone, PostgreSQL, PGVector, OpenAI Embeddings, LLM-to-SQL

Deployment & Cloud Infrastructure
Node.js / TypeScript, Docker, AWS, DigitalOcean, Linux VPS, Render Web Hosting, GitHub CI/CD

Integration & Orchestration
MCP Client, REST APIs, Webhooks, Schedulers, OAuth2, Third-Party APIs

Business & CRM Integrations
HubSpot CRM, GoHighLevel (GHL), Airtable, Microsoft 365, Slack, Lark Suite, Teamwork, ConnectWise

Social & Messaging Platforms
WhatsApp Cloud API / WAPA, Instagram DM Automations, Facebook Messenger, Telegram Bot API

Forms & Lead Capture Pipelines
Typeform, Tally, Fillout, Jotform, Google Forms, Lead Validation & Multi-Step Conditional Routing

Selected Project Highlights

AI MicroMind App & Reporting Bot Framework
Conversational BI reporting engine translating natural language prompts into live SQL/OData queries across Oracle, SAP, Dynamics 365, Odoo, and SQL databases.

30+ Native Integration Nodes (AI MicroMind Core)
Reusable native integration nodes for WAPA (WhatsApp API), Telegram, Gmail, MCP Client, Pinecone, webhooks, schedulers, and PostgreSQL audit logging.

Smile Inn Dental Clinic Website (Live Concept Mockup — Render)
Designed a modern, responsive clinic website concept mockup featuring calming color psychology, interactive before/after treatment sliders, doctor profiles, and appointment booking funnels.

FOE Sovereign Procurement Directorate Portal (Concept Prototype — Render)
Conceptual digital web portal and dashboard mockup designed for sovereign procurement operations with executive branding, responsive data tables, multi-role navigation, and bilingual Arabic/English layout.

Omnichannel Social & Messaging Automation Systems
Multi-platform social funnels connecting WhatsApp Cloud API / WAPA, Instagram DM comment-to-lead triggers, Facebook Messenger support bots, and Telegram channels with HubSpot & GoHighLevel.

Smart Task Assistant & Merna Invitation Web Apps (Live Prototypes — Render)
AI-powered task management app prototype with sleek dark-mode glassmorphism alongside a bespoke mobile-first animated digital invitation web mockup.